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```
openExample('econ/VARModelCaseStudyExample')
```



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VAR Model Case Study

This example shows how to analyze a VAR model.

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Overview of Case Study

This section contains an example of the workflow described in [docid:econ_ug.bsxcdon-1](#). The example uses three time series: GDP, M1 money supply, and the 3-month T-bill rate. The example shows:

- Loading and transforming the data for stationarity
- Partitioning the transformed data into presample, estimation, and forecast intervals to support a backtesting experiment
- Making several models
- Fitting the models to the data
- Deciding which of the models is best
- Making forecasts based on the best model

Load and Transform Data

The file `Data_USEconModel` ships with Econometrics Toolbox™ software. The file contains time series from the Federal Reserve Bank of St. Louis Economics Data (FRED) database in a tabular array. This example uses three of the time series:

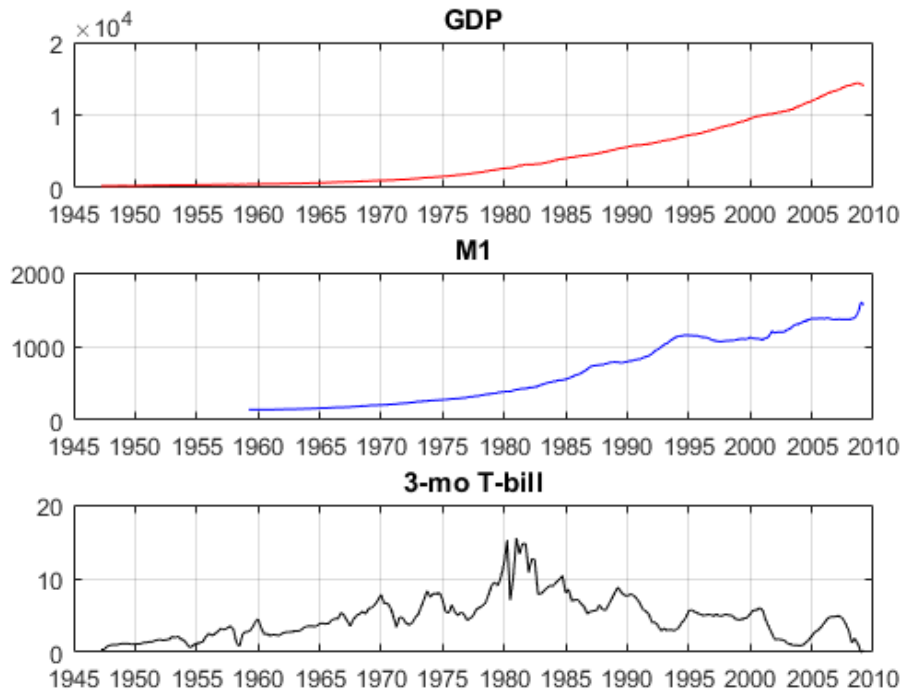
- GDP (GDP)
- M1 money supply (M1SL)
- 3-month T-bill rate (TB3MS)

Load the data into a time series matrix `Y`.

```
load Data_USEconModel
gdp = DataTable.GDP;
m1 = DataTable.M1SL;
tb3 = DataTable.TB3MS;
Y = [gdp,m1,tb3];
```

Plot the data to look for trends.

```
figure
subplot(3,1,1)
plot(dates,Y(:,1),'r');
title('GDP')
datetick('x')
grid on
subplot(3,1,2);
plot(dates,Y(:,2),'b');
title('M1')
datetick('x')
grid on
subplot(3,1,3);
plot(dates, Y(:,3), 'k')
title('3-mo T-bill')
datetick('x')
grid on
hold off
```



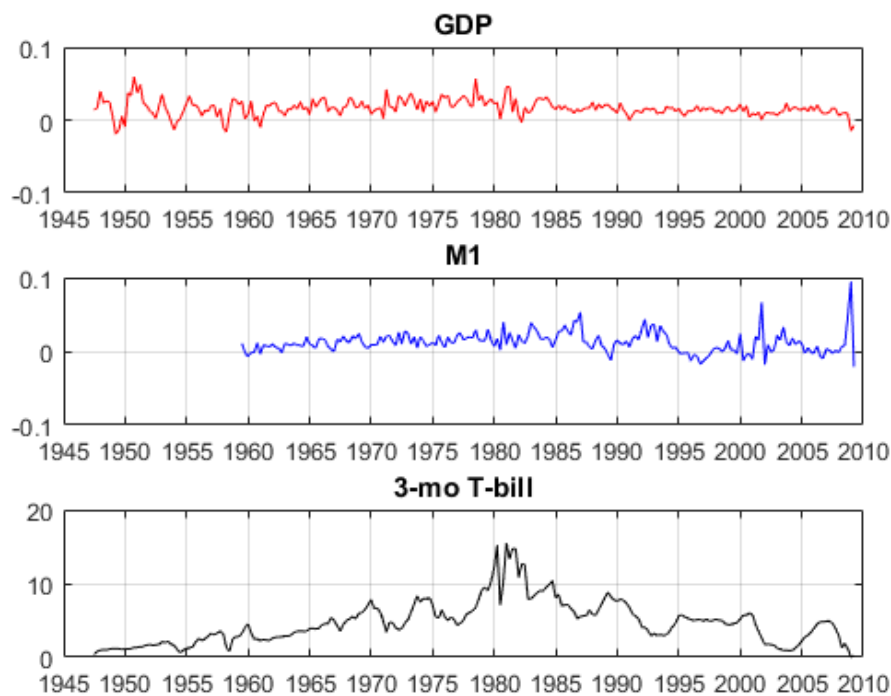
The GDP and M1 data appear to grow, while the T-bill returns show no long-term growth. To counter the trends in GDP and M1, take a difference of the logarithms of the data. Taking a difference shortens the time series. Therefore, truncate the T-bill series and date series X so that the Y data matrix has the same number of rows for each column.

```

Y = [diff(log(Y(:,1:2))), Y(2:end,3)]; % Transformed data
X = dates(2:end);

figure
subplot(3,1,1)
plot(X,Y(:,1),'r');
title('GDP')
datetick('x')
grid on
subplot(3,1,2);
plot(X,Y(:,2),'b');
title('M1')
datetick('x')
grid on
subplot(3,1,3);
plot(X, Y(:,3), 'k'),
title('3-mo T-bill')
datetick('x')
grid on

```

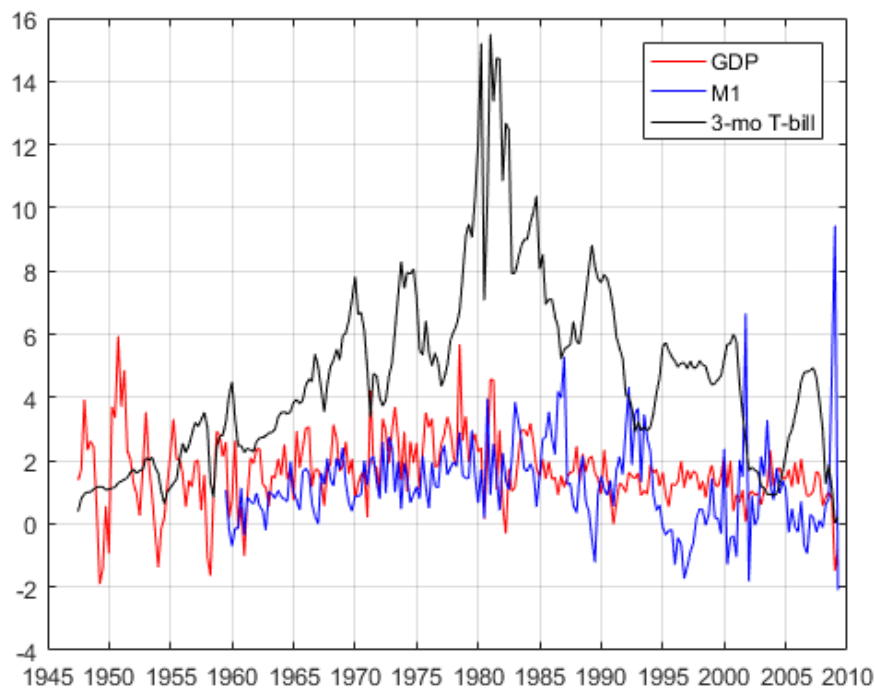


The scale of the first two columns is about 100 times smaller than the third. Multiply the first two columns by 100 so that the time series are all roughly on the same scale. This scaling makes it easy to plot all the series on the same plot. More importantly, this type of scaling makes optimizations more numerically stable (for example, maximizing loglikelihoods).

```

Y(:,1:2) = 100*Y(:,1:2);
figure
plot(X,Y(:,1),'r');
hold on
plot(X,Y(:,2),'b');
datetick('x')
grid on
plot(X,Y(:,3),'k');
legend('GDP','M1','3-mo T-bill');
hold off

```



Select and Fit the Models

You can select many different models for the data. This example uses four models.

- VAR(2) with diagonal autoregressive and covariance matrices
- VAR(2) with full autoregressive and covariance matrices
- VAR(4) with diagonal autoregressive and covariance matrices
- VAR(4) with full autoregressive and covariance matrices

Make all the series the same length, and transform them to be stationary and on a similar scale.

```
dGDP = 100*diff(log(gdp(49:end)));
dM1 = 100*diff(log(m1(49:end)));
dT3 = diff(tb3(49:end));
Y = [dGDP dM1 dT3];
```

Create the four models.

```
dt = logical(eye(3));
VAR2diag = vgxset('ARsolve', repmat({dt},2,1),...
    'asolve',true(3,1),'Series',{'GDP','M1','3-mo T-bill'});
VAR2full = vgxset(VAR2diag,'ARsolve',[]);
VAR4diag = vgxset(VAR2diag,'nAR',4,'ARsolve',repmat({dt},4,1));
VAR4full = vgxset(VAR2full,'nAR',4);
```

The matrix `dt` is a diagonal logical matrix. `dt` specifies that both the autoregressive matrices for `VAR2diag` and `VAR4diag` are diagonal. In contrast, the specifications for `VAR2full` and `VAR4full` have empty matrices instead of `dt`. Therefore, `vgxvarx` fits the defaults, which are full matrices for autoregressive and correlation matrices.

To assess the quality of the models, divide the response data `Y` into three periods: presample, estimation, and forecast. Fit the models to the estimation data, using the presample period to provide lagged data. Compare the predictions of the fitted models to the forecast data. The estimation period is in sample, and the forecast period is out of sample (also known

as backtesting).

For the two VAR(4) models, the presample period is the first four rows of Y . Use the same presample period for the VAR(2) models so that all the models are fit to the same data. This is necessary for model fit comparisons. For both models, the forecast period is the final 10% of the rows of Y . The estimation period for the models goes from row 5 to the 90% row. Define these data periods.

```
YPre = Y(1:4,:);
T = ceil(.9*size(Y,1));
YEst = Y(5:T,:);
YF = Y((T+1):end,:);
TF = size(YF,1);
```

Now that the models and time series exist, you can easily fit the models to the data.

```
[EstSpec1,EstStdErrors1,logL1,W1] = ...
    vgxvarx(VAR2diag,YEst,[],YPre,'CovarParams','Diagonal');
[EstSpec2,EstStdErrors2,logL2,W2] = ...
    vgxvarx(VAR2full,YEst,[],YPre);
[EstSpec3,EstStdErrors3,logL3,W3] = ...
    vgxvarx(VAR4diag,YEst,[],YPre,'CovarParams','Diagonal');
[EstSpec4,EstStdErrors4,logL4,W4] = ...
    vgxvarx(VAR4full,YEst,[],YPre);
```

- The EstSpec structures are the fitted models.
- The EstStdErrors structures contain the standard errors of the fitted models.
- The logL values are the loglikelihoods of the fitted models, which you use to help select the best model.
- The W vectors are the estimated innovations (residuals) processes, which are the same size as YEst.
- The specification for EstSpec1 and EstSpec3 includes diagonal covariance matrices.

Check Model Adequacy

You can check whether the estimated models are stable and invertible using the `vgxqual` function. (There are no MA terms in these models, so the models are necessarily invertible.) The test shows that all the estimated models are stable.

```
[isStable1,isInvertible1] = vgxqual(EstSpec1);
[isStable2,isInvertible2] = vgxqual(EstSpec2);
[isStable3,isInvertible3] = vgxqual(EstSpec3);
[isStable4,isInvertible4] = vgxqual(EstSpec4);
[isStable1,isStable2,isStable3,isStable4]
```

ans =

1×4 logical array

1 1 1 1

You can also look at the estimated specification structures. Each contains a line stating whether the model is stable.

```
EstSpec4
```

```
EstSpec4 =
```

```
Model: 3-D VAR(4) with Additive Constant
Series: {'GDP' 'M1' '3-mo T-bill'}
n: 3
nAR: 4
nMA: 0
nX: 0
a: [0.524224 0.106746 -0.671714] additive constants
asolve: [1 1 1 logical] additive constant indicators
AR: {4x1 cell} stable autoregressive process
ARsolve: {4x1 cell of logicals} autoregressive lag indicators
Q: [3x3] covariance matrix
Qsolve: [3x3 logical] covariance matrix indicators
```

AR: {4x1 cell} stable autoregressive process appears in the output indicating that the autoregressive process is stable.

You can compare the restricted (diagonal) AR models to their unrestricted (full) counterparts using `lratiotest`. The test rejects or fails to reject the hypothesis that the restricted models are adequate, with a default 5% tolerance. This is an in-sample test.

Apply the likelihood ratio tests by counting the parameters in the models using `vgxcount`, and then use `lratiotest` to perform the tests.

```
[n1,n1p] = vgxcount(EstSpec1);
[n2,n2p] = vgxcount(EstSpec2);
[n3,n3p] = vgxcount(EstSpec3);
[n4,n4p] = vgxcount(EstSpec4);
reject1 = lratiotest(logL2,logL1,n2p - n1p)
reject3 = lratiotest(logL4,logL3,n4p - n3p)
reject4 = lratiotest(logL4,logL2,n4p - n2p)
```

```
reject1 =
```

```
logical
```

```
1
```

```
reject3 =
```

```
logical
```

```
1
```

```
reject4 =
```

```
logical
```

```
0
```

The 1 results indicate that the likelihood ratio test rejected both the restricted models, AR(1) and AR(3), in favor of the corresponding unrestricted models. Therefore, based on this test, the unrestricted AR(2) and AR(4) models are preferable. However, the test does not reject the unrestricted AR(2) model in favor of the unrestricted AR(4) model. (This test regards the AR(2) model as an AR(4) model with restrictions that the autoregression matrices AR(3) and AR(4) are 0.) Therefore, it seems that the unrestricted AR(2) model is the best model.

To find the best model in a set, minimize the Akaike information criterion (AIC). Use in-sample data to compute the AIC. Calculate the criterion for the four models.

```
AIC = aicbic([logL1 logL2 logL3 logL4],[n1p n2p n3p n4p])
```

AIC =

1.0e+03 *

1.5129 1.4462 1.5122 1.4628

The best model according to this criterion is the unrestricted AR(2) model. Notice, too, that the unrestricted AR(4) model has lower Akaike information than either of the restricted models. Based on this criterion, the unrestricted AR(2) model is best, with the unrestricted AR(4) model coming next in preference.

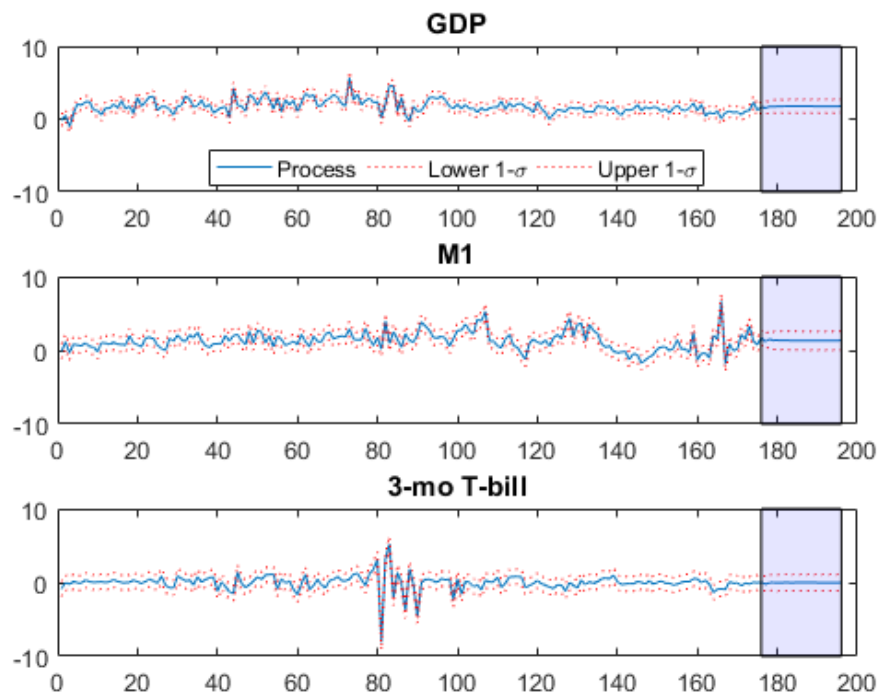
To compare the predictions of the four models against the forecast data YF, use `vgxpred`. This function returns both a prediction of the mean time series, and an error covariance matrix that gives confidence intervals about the means. This is an out-of-sample calculation.

```
[FY1,FYCov1] = vgxpred(EstSpec1,TF,[],YEst);
[FY2,FYCov2] = vgxpred(EstSpec2,TF,[],YEst);
[FY3,FYCov3] = vgxpred(EstSpec3,TF,[],YEst);
[FY4,FYCov4] = vgxpred(EstSpec4,TF,[],YEst);
```

This plot shows the predictions in the shaded region to the right.

```
figure
vgxplot(EstSpec2,YEst,FY2,FYCov2)
```

MATLAB Examples



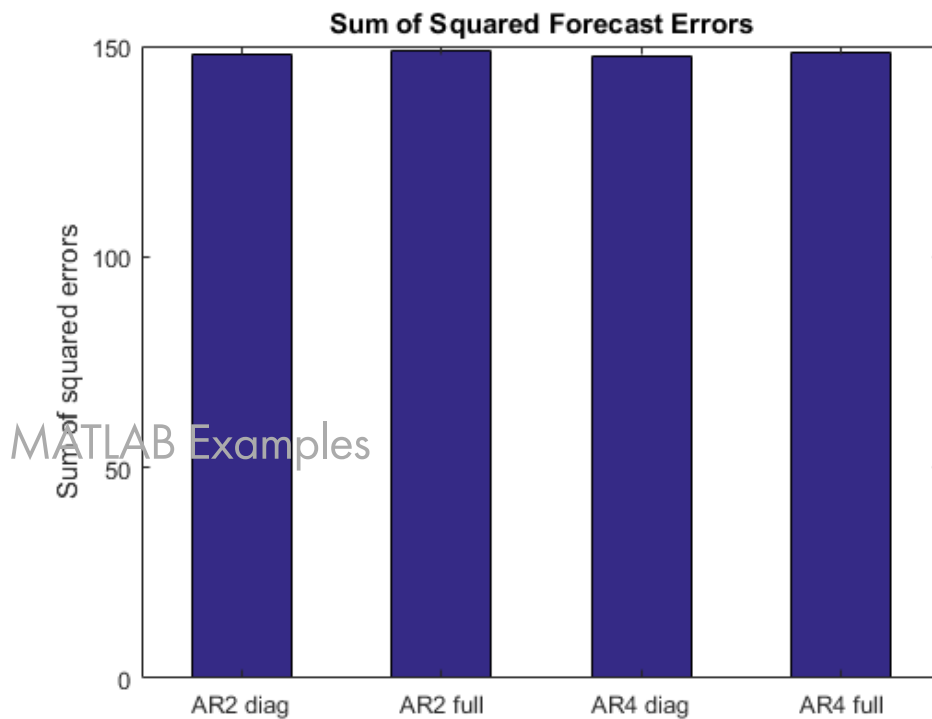
It is now straightforward to calculate the sum-of-squares error between the predictions and the data, YF.

```

Error1 = YF - FY1;
Error2 = YF - FY2;
Error3 = YF - FY3;
Error4 = YF - FY4;

SSerror1 = Error1(:)' * Error1(:);
SSerror2 = Error2(:)' * Error2(:);
SSerror3 = Error3(:)' * Error3(:);
SSerror4 = Error4(:)' * Error4(:);
figure
bar([SSerror1 SSerror2 SSerror3 SSerror4],.5)
ylabel('Sum of squared errors')
set(gca,'XTickLabel',...
    {'AR2 diag' 'AR2 full' 'AR4 diag' 'AR4 full'})
title('Sum of Squared Forecast Errors')

```



The predictive performances of the four models are similar.

The full AR(2) model seems to be the best and most parsimonious fit. Its model parameters are as follows.

```
vgxdisp(EstSpec2)
```



```

Model : 3-D VAR(2) with Additive Constant
        Conditional mean is AR-stable and is MA-invertible
Series : GDP
Series : M1
Series : 3-mo T-bill
a Constant:
    0.687401
    0.3856
   -0.915879
AR(1) Autoregression Matrix:
    0.272374   -0.0162214   0.0928186
    0.0563884   0.240527   -0.389905
    0.280759   -0.0712716   -0.32747
AR(2) Autoregression Matrix:
    0.242554   0.140464   -0.177957
    0.00130726  0.380042   -0.0484981
    0.260414   0.024308   -0.43541
Q Innovations Covariance:
    0.632182   0.105925   0.216806
    0.105925   0.991607   -0.155881
    0.216806   -0.155881   1.00082

```

Forecast Observations

You can make predictions or forecasts using the fitted model (EstSpec4) either by:

- Running `vgxpred` based on the last few rows of `YF`
- Simulating several time series with `vgxsim`

In both cases, transform the forecasts so they are directly comparable to the original time series.

Generate 10 predictions from the fitted model beginning at the latest times using `vgxpred`.

```
[YPred,YCov] = vgxpred(EstSpec2,10,[],YF);
```

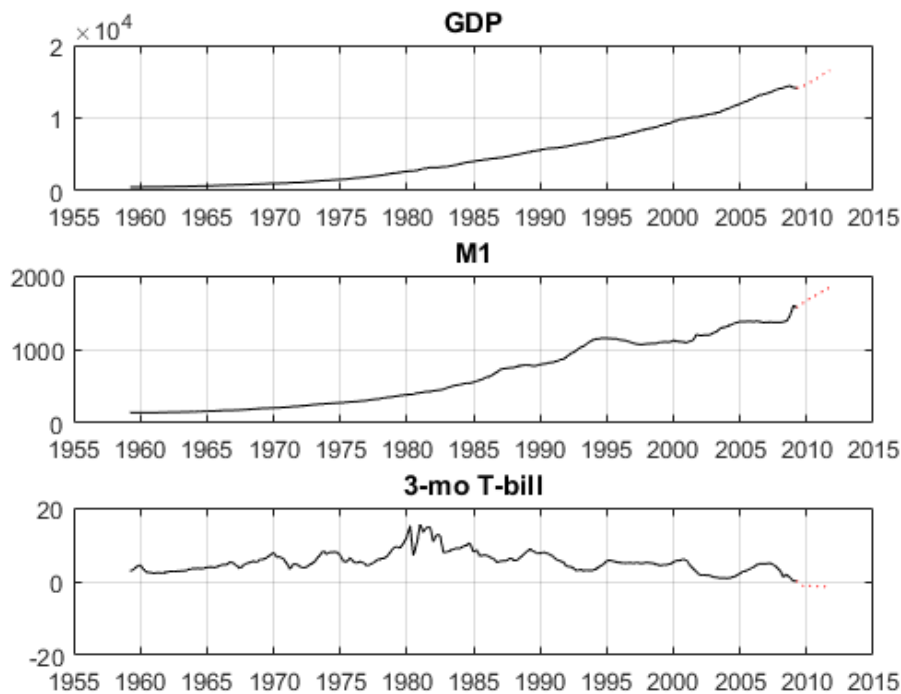
Transform the predictions by undoing the scaling and differencing applied to the original data. Make sure to insert the last observation at the beginning of the time series before using `cumsum` to undo the differencing. And, since differencing occurred after taking logarithms, insert the logarithm before using `cumsum`.

```

YFirst = [gdp,m1,tb3];
YFirst = YFirst(49:end,:);           % Remove NaNs
dates = dates(49:end);
EndPt = YFirst(end,:);
EndPt(1:2) = log(EndPt(1:2));
YPred(:,1:2) = YPred(:,1:2)/100;     % Rescale percentage
YPred = [EndPt; YPred];              % Prepare for cumsum
YPred(:,1:3) = cumsum(YPred(:,1:3));
YPred(:,1:2) = exp(YPred(:,1:2));
lasttime = dates(end);
timess = lasttime:91:lasttime+910;   % Insert forecast horizon

figure
subplot(3,1,1)
plot(timess,YPred(:,1),'r')
hold on
plot(dates,YFirst(:,1),'k')
datetick('x')
grid on
title('GDP')
subplot(3,1,2);
plot(timess,YPred(:,2),'r')
hold on
plot(dates,YFirst(:,2),'k')
datetick('x')
grid on
title('M1')
subplot(3,1,3);
plot(timess,YPred(:,3),'r')
hold on
plot(dates,YFirst(:,3),'k')
datetick('x')
grid on
title('3-mo T-bill')
hold off

```



The plots show the extrapolations in dotted red, and the original data series in solid black.

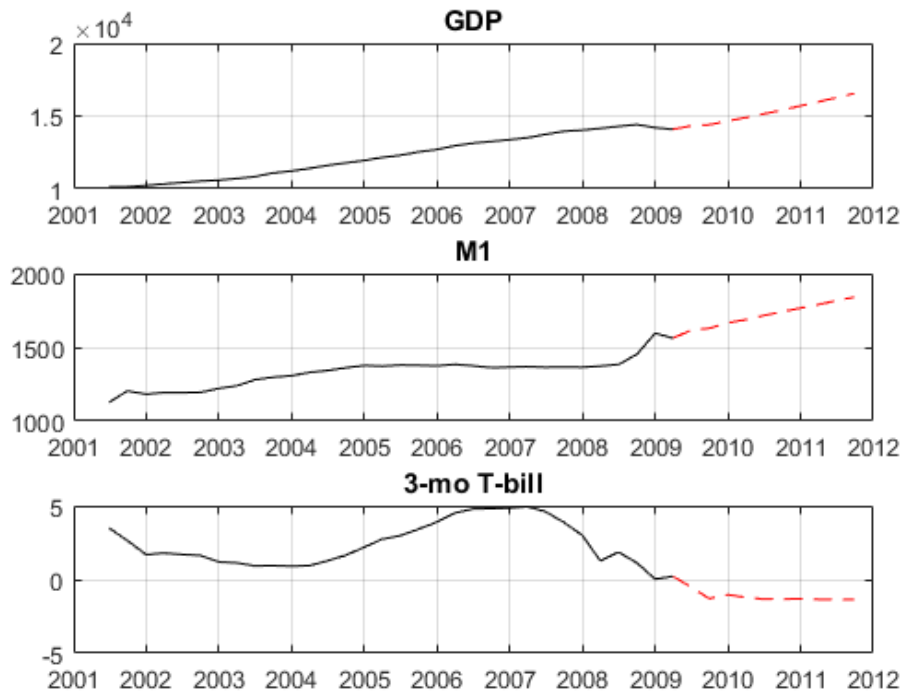
Look at the last few years in this plot to get a sense of how the predictions relate to the latest data points.

```

Ylast = YFirst(170:end,:);
timeslast = dates(170:end);

figure
subplot(3,1,1)
plot(times,YPred(:,1),'--r')
hold on
plot(timeslast,Ylast(:,1),'k')
datetick('x')
grid on
title('GDP')
subplot(3,1,2);
plot(times,YPred(:,2),'--r')
hold on
plot(timeslast,Ylast(:,2),'k')
datetick('x')
grid on
title('M1')
subplot(3,1,3);
plot(times,YPred(:,3),'--r')
hold on
plot(timeslast,Ylast(:,3),'k')
datetick('x')
grid on
title('3-mo T-bill')
hold off

```



The forecast shows increasing GDP, little growth in M1, and a slight decline in the interest rate. However, the forecast has no error bars.

Alternatively, you can generate 10 predictions from the fitted model beginning at the latest times using `vgxsim`. This method simulates 2000 time series times, and then generates the means and standard deviations for each period. The means of the deviates for each period are the predictions for that period.

Simulate a time series from the fitted model beginning at the latest times.

```
rng(1); % For reproducibility
YSim = vgxsim(EstSpec2,10,[],YF,[],2000);
```

Transform the predictions by undoing the scaling and differencing applied to the original data. Make sure to insert the last observation at the beginning of the time series before using `cumsum` to undo the differencing. And, since differencing occurred after taking logarithms, insert the logarithm before using `cumsum`.

```
YFirst = [gdp,m1,tb3];
EndPt = YFirst(end,:);
EndPt(1:2) = log(EndPt(1:2));
YSim(:,1:2,:) = YSim(:,1:2,)/100;
YSim = [repmat(EndPt,[1,1,2000]);YSim];
YSim(:,1:3,:) = cumsum(YSim(:,1:3,:));
YSim(:,1:2,:) = exp(YSim(:,1:2,:));
```

Compute the mean and standard deviation of each series, and plot the results. The plot has the mean in black, with a ± 1 standard deviation in red.

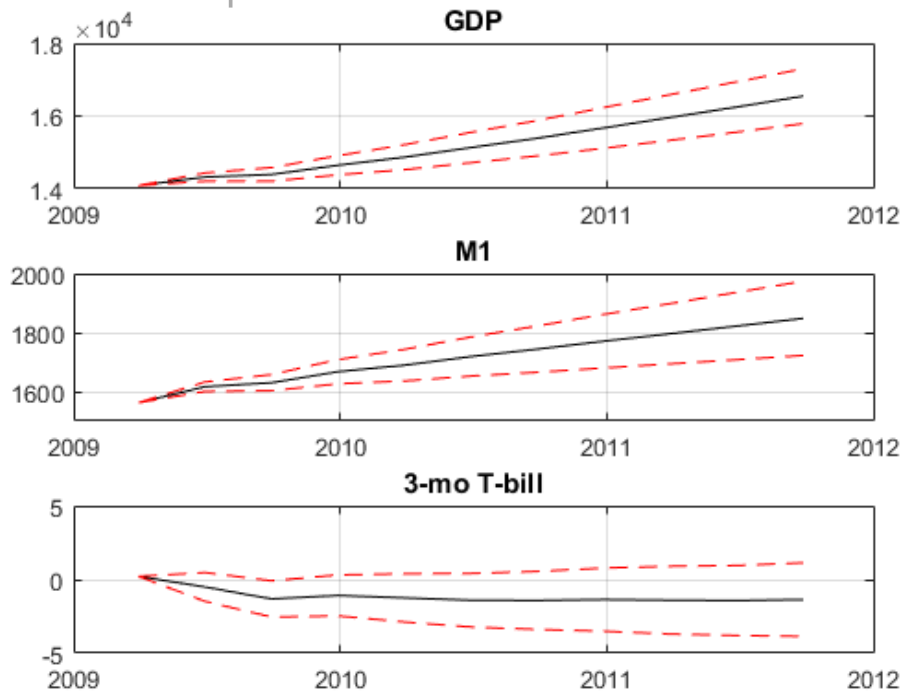
```

YMean = mean(YSim,3);
YSTD = std(YSim,0,3);

figure
subplot(3,1,1)
plot(timess,YMean(:,1),'k')
datetick('x')
grid on
hold on
plot(timess,YMean(:,1)+YSTD(:,1),'--r')
plot(timess,YMean(:,1)-YSTD(:,1),'--r')
title('GDP')
subplot(3,1,2);
plot(timess,YMean(:,2),'k')
hold on
datetick('x')
grid on
plot(timess,YMean(:,2)+YSTD(:,2),'--r')
plot(timess,YMean(:,2)-YSTD(:,2),'--r')
title('M1')
subplot(3,1,3);
plot(timess,YMean(:,3),'k')
hold on
datetick('x')
grid on
plot(timess,YMean(:,3)+YSTD(:,3),'--r')
plot(timess,YMean(:,3)-YSTD(:,3),'--r')
title('3-mo T-bill')
hold off

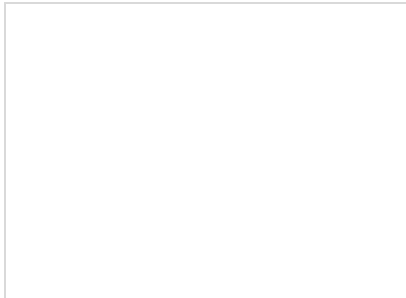
```

MATLAB Examples



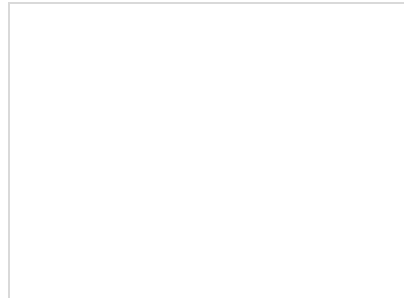
The plots show increasing growth in GDP, moderate to little growth in M1, and uncertainty about the direction of T-bill rates.

Related Examples



Box-Jenkins Model Selection

This example shows how to use the Box-Jenkins methodology to select an ARIMA model. The time series is the log quarterly Australian Consumer Price



Forecast a VAR Model Using Monte Carlo Simulation

This example shows how to use Monte Carlo simulation via `vgxsim` to forecast a VAR model. Multivariate Models
Econometrics Toolbox Computational

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MATLAB Examples